



Scraping Metrics in Prometheus

Prometheus collects and stores metrics using a pull-based model. It scrapes metrics from configured targets at specified intervals.

Why Scrape Metrics?

- Collect real-time performance data
 - Monitor application health
 - Identify system bottlenecks
 - Automate alerts and monitoring
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How Prometheus Scrapes Metrics

- 1 **Prometheus pulls data** from **HTTP endpoints** (Exposed by applications).
 - 2 Applications expose metrics in **OpenMetrics format** (a text-based format).
 - 3 **Prometheus stores scraped metrics** in a time-series database.
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Configuring Prometheus for Scraping

Step 1: Define Targets in **prometheus.yml**

Edit the Prometheus configuration file:

```
global:
  scrape_interval: 15s # Scrape every 15 seconds

scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
```



```
- targets: ['localhost:9100'] # Node Exporter target
```

Step 2: Start Prometheus

```
./prometheus --config.file=prometheus.yml
```

Step 3: Verify Scraped Metrics

Visit <http://localhost:9090/targets> to check target health.

Visit <http://localhost:9100/metrics> to see scraped metrics.

Know More

Q1: What happens if a target is unreachable?

Prometheus **logs an error** and retries scraping in the next interval.

Q2: Can Prometheus scrape metrics from multiple sources?

Yes! Define multiple **scrape_configs** in **prometheus.yml**.

Now you can configure Prometheus to scrape metrics and monitor applications!